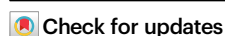


Wafer-scale high- κ HfO₂ dielectric films with sub-5-Å equivalent oxide thickness for 2D MoS₂ transistors

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High- κ gate dielectrics are indispensable in modern transistor technology and play a pivotal role in efficient capacitive gating and suppression of leakage currents. However, the realization of industry-compatible high- κ gate dielectrics at a sub-5-Å equivalent oxide thickness (EOT) remains challenging. Here we report the realization of 1.3-nm thick hafnium oxide (HfO₂) dielectrics via an industry-compatible multiple oxidation atomic layer deposition process at 200 °C. A low EOT down to 2.5 Å is demonstrated for 1.3-nm thick HfO₂ dielectrics on metal gates with a low leakage current of 10⁻⁶ A/cm² and a robust breakdown electric field of ~22.3 MV/cm. Remarkably, such low EOT high- κ /metal gates can be directly implanted into emerging two-dimensional (2D) transistors and low-power logic circuits on 8-inch wafer scale to showcase their potentials. The as-fabricated molybdenum disulfide (MoS₂) transistors exhibit a large on-state current density of 260 μ A/ μ m at source-drain bias of 0.5 V, a high on/off ratio of 10⁸, an average subthreshold slope (SS) of 75 mV/dec, and small capacitance equivalent thickness (CET) values of 0.34 nm for gate-first transistors and 0.50 nm for gate-last transistors. Our ultra-scaled dielectrics hold significant promise for advanced semiconductor fabrication processes towards the angstrom era.

The evolution of semiconductor technology is closely associated with the development of high- κ gate dielectrics, such as aluminum oxide (Al₂O₃), hafnium oxide (HfO₂), and zirconium oxide (ZrO₂)^{1–4}. In 2007, Intel Corporation announced the adoption of hafnium-based high- κ

metal gate (HKMG) in its 45 nm technology node. This technological innovation marked a significant paradigm shift away from traditional silicon dioxide (SiO₂) dielectrics and poly-Si gate, quickly becoming a standard in the semiconductor industry. In order to further advance

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semiconductor technology nodes for more compact, fast, and energy-efficient integrated circuits (ICs)^{5–8}, it is pivotal to continue the equivalent oxide thickness (EOT) scaling of high- κ dielectrics in transistors. EOT of a high- κ dielectric refers to the equivalent thickness of a SiO₂ film having the same specific capacitance with the high- κ dielectric film, while capacitance equivalent thickness (CET) reflects the actual physical gate controllability in devices. The former is more commonly used to evaluate the dielectric performance of one material. In the process of EOT reduction, using high- κ materials can effectively suppress the leakage current. EOT within silicon transistors at the 3-nm technology node has already been reduced to 9.5 Å⁹. Recently, high- κ dielectrics of HfO₂-ZrO₂ superlattice or Sb₂O₃/HfO₂ with EOT of 6–8 Å has been demonstrated in prototype transistors^{10,11}.

According to the 2022 International Roadmap for Devices and Systems (IRDS), the outlined EOT for emerging logic devices requires to be <5 Å¹². Such aggressive EOT values are challenging to achieve in conventional HfO₂- or ZrO₂-based high- κ dielectrics^{10–14}. Over the past five years, many emerging high- κ dielectrics, including perovskite SrTiO₃, ion crystal CaF₂, molecular crystal Sb₂O₃, and two-dimensional (2D) Bi₂SeO₃/Bi₂SiO₅, have been developed for further EOT scaling down to ~4.1 Å^{15–19}. Yet promising, these materials are incompatible with the standard manufacturing process line due to high-temperature processes, small-scale limitations, and specific channel necessities, etc.²⁰. Besides, these emerging high- κ dielectrics also exhibit large gate leakage currents at 10^{–1}–10^{–3} A/cm² level, which are away from the low-power application of 10^{–5} A/cm²^{11,13,14}. Thus, developing a high- κ gate dielectric with sub-5-Å EOT, low gate leakage currents of <10^{–5} A/cm², and implementable to standard semiconductor manufacturing process as well persists as an essential yet elusive challenge.

In this work, we develop a multiple oxidation atomic layer deposition (MOALD) approach for high-quality ultrathin HfO₂ dielectrics at an 8-inch wafer scale. The deposition is performed at a relatively low temperature of 200 °C, rendering it compatible not only with commercial Si complementary metal-oxide-semiconductor (CMOS) technology but also with emerging two-dimensional semiconductor-based field-effect transistors (2D FETs). Notably, we report the realization of the 1.3-nm-thick HfO₂ dielectric layer with a small EOT of 2.5 Å. As-achieved HfO₂ films exhibit an exceedingly low leakage current below 10^{–6} A/cm² at 1 V gate voltage and showcase a robust breakdown electric field of ~22.3 MV/cm. The application of such a low-EOT high- κ layer in cutting-edge 2D gate-first and gate-last FETs based on monolayer molybdenum disulfide (MoS₂) is demonstrated to evaluate its practicality and efficacy in gate control. Remarkably, 100-nm-channel devices feature an on-state current density of ~260 μA/μm, a high on/off ratio of ~10⁸, and an average subthreshold slope (SS) of ~75 mV/dec at the source-drain bias of 0.5 V. We further fabricate 8-inch wafer-scale logic gates and memory devices such as inverters, logic gates (e.g., AND, NAND, and NOR) and five-stage ring oscillators (ROs) with correct functions. Such ultimately scaled EOT not only transcends the goal of IRDS for 2037, but also holds substantial implications for the forthcoming advanced semiconductor processes towards the ångström era.

Results

Realization of the 2.5-Å-EOT HfO₂ films

In the conventional ALD process for HfO₂ employing only one oxidation step, a wealth of oxygen vacancies (O_v) is often found in the as-deposited films. In contrast, our optimized HfO₂ deposition is performed via a MOALD process, including two steps of oxidation by ozone and oxygen plasma accordingly. Please refer to Materials and Methods for more details. As illustrated in Fig. 1a, this MOALD process is effective in reducing oxygen vacancy concentrations (V_O) and significantly enhancing the quality of HfO₂ films, as confirmed from the X-ray photoelectron spectroscopy (XPS) spectra (Fig. 1b). From the XPS characterizations, it can be clearly seen that V_O reduces from 25.7%

for one-step-oxidation ALD to nearly negligible levels for MOALD²¹. As a result, when applying in MoS₂ FETs, hysteresis in the transfer curve is expected to be greatly suppressed with MOALD HfO₂ dielectric, compared with that using normal ALD HfO₂ dielectrics. (Supplementary Fig. 1).

We further elucidate the merit of the reduced defect concentration in HfO₂ film through first-principles calculations (Supplementary Fig. 2). V_O is defined as:

$$V_O = \frac{n}{N} \times 100\%$$

where N equals to 320, representing the total number of oxygen atoms; and n is the number of O_v. Here, we consider several different V_O varying from 0 to 15.6% (Supplementary Fig. 3). Five amorphous structures for each V_O are generated and optimized with fixed lattice parameters (Supplementary Fig. 3). The densities of states at each V_O are calculated and shown in Supplementary Fig. 4. The Fig. 1c, d show results from two typical V_O, i.e., 0 and 15.6%, respectively. Due to the presence of structural disorder, the electrical conductivities of the systems are assessed through the mobility bandgap, which is identified using the inverse participation ratio^{22–24} (see Supplementary Notes 1, 2 and Supplementary Fig. 5). The mobility bandgaps are summarized in Supplementary Tables 1–6. The average mobility bandgap at each V_O is presented in Fig. 1e, which shows a shrinking trend with the increase of V_O. With the mobility bandgap reducing at the presence of oxygen vacancies, a reduced breakdown strength of HfO₂ is expected. Moreover, as V_O increases, in-gap localized defect states become more prominent (Fig. 1d and Supplementary Fig. 4). These defective states are expected to provide additional channels, like trap assisted tunneling (TAT), for electron tunneling thus further reduces the breakdown strength of HfO₂. Therefore, reducing the O_v concentration is expected to restore the breakdown strength of HfO₂ thin film and enhance its dielectric performance.

In order to fabricate metal/high- κ stacks, we first deposit 10-nm Au at a slow rate of 0.01 Å/s by the electron-beam evaporation then HfO₂ by the MOALD. At least 10 ALD cycles are necessary to realize continuous HfO₂ films with a minimum thickness of 1.3 nm. Figure 1f and Supplementary Fig. 6 show the cross-sectional transmission electron microscopy (CS-TEM) images of an Au/HfO₂/MoS₂ stack, demonstrating the continuity and uniformity of the 1.3-nm HfO₂ film. A zoomed-in image is shown in Fig. 1g with two stacked layers of monolayer MoS₂ and HfO₂ clearly seen. The layer thickness of MoS₂ and HfO₂ are 0.7 and 1.3 nm, respectively. Besides, the stacked structure of Au/HfO₂/MoS₂ is further confirmed from the elemental mapping of Au, Mo, Hf (Fig. 1h). Atomic force microscopy (AFM) images on the surface of gate metal before and after deposition of 1.3-nm-HfO₂ layer are also shown in Fig. 1i, indicating an excellent surface smoothness, integrity, and uniformity. The surface roughness of the metal gate and dielectric are 241 and 239 pm, respectively, as illustrated in the height histogram (Fig. 1j and Supplementary Fig. 7).

Dielectric properties of MOALD ultrathin HfO₂ films

The dielectric properties of MOALD HfO₂ dielectrics are characterized electrically in a metal-insulator-metal (MIM) cell, in which employing Au as top and bottom electrode. (Fig. 2a, b) The C-V and leakage-current measurements are performed for 1.3-, 2.6- and 3.9-nm thick HfO₂ films (Fig. 2c and Supplementary Fig. 8a). Our results indicate that κ is ~31.0 for 3.9-nm-HfO₂ films; and it gradually reduces with the decreasing layer thickness but still maintains at 19.8 for 1.3-nm-HfO₂ dielectrics (Fig. 2d). As EOT = 3.9*t/ κ , where t denotes the dielectric thickness, we also plot the corresponding EOT for different HfO₂ thickness in Fig. 2d. Notably, an EOT of 2.5 Å is achieved for 1.3-nm-HfO₂ dielectrics.

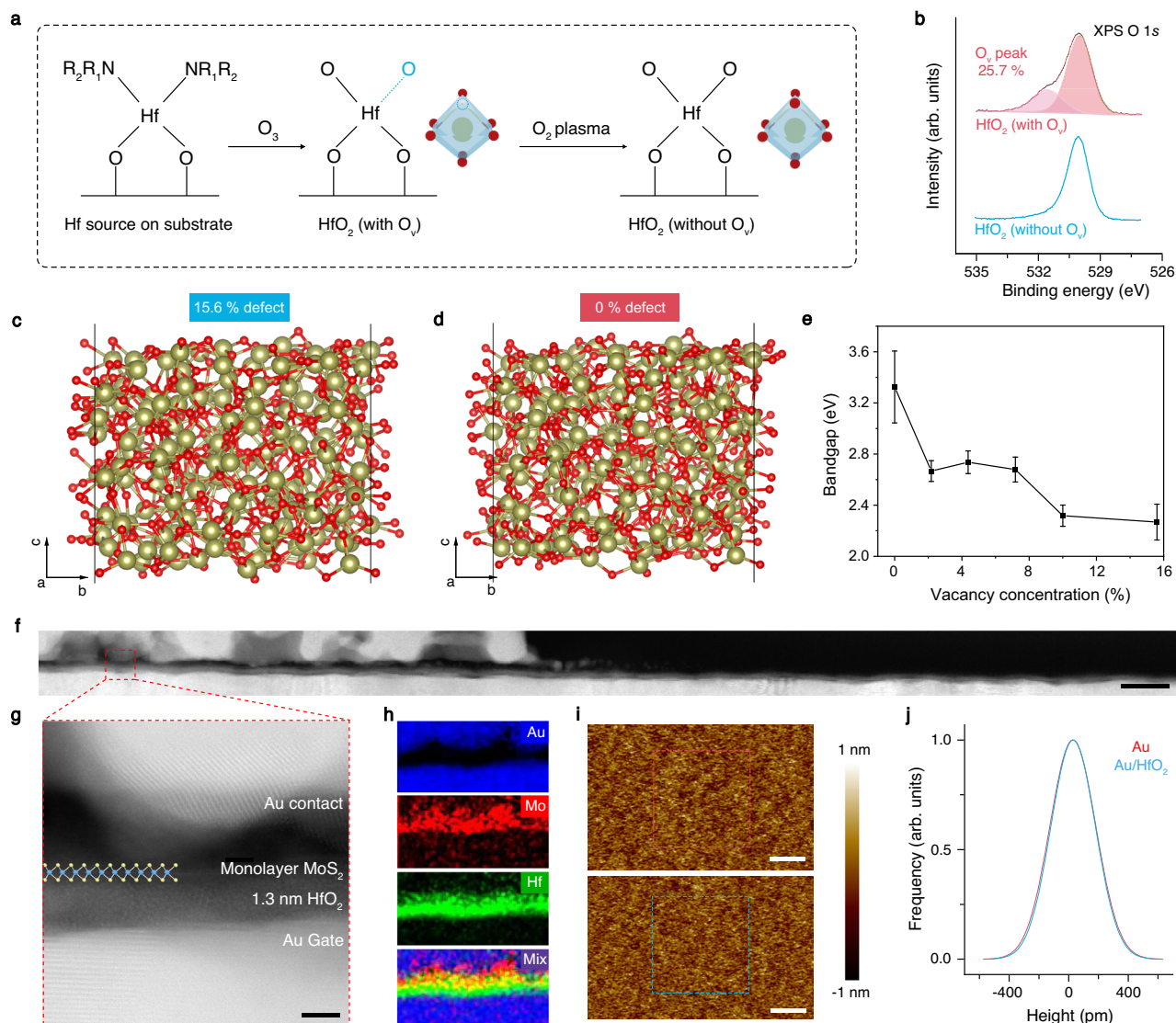


Fig. 1 | Multiple oxidation atomic layer deposition (MOALD) of 2.5 Å equivalent oxide thickness (EOT) HfO₂ films. **a** Schematic illustration of HfO₂ with and without O_v. **b** The X-ray photoelectron spectroscopy (XPS) spectrum of O 1s in HfO₂ with and without O_v. The structures of HfO₂ slabs with **c** 15.6% and **d** 0% O_v defect concentrations calculated by Perdew–Burke–Ernzerhof (PBE) functional. **e** The average bandgaps of HfO₂ slabs with different O_v defect concentrations. The error bar is calculated with standard deviation. **f** Cross-sectional transmission

electron microscopy (CS-TEM) image of the Au/HfO₂/MoS₂ stack. Scale bar: 50 nm. **g** Enlarged CS-TEM image of the stacking layers. Scale bar: 1 nm. **h** Scanning TEM energy dispersive X-ray spectroscopy (STEM-EDS) elemental mapping of S, Hf, and Ti, respectively, acquired from the same area in **(g)**. **i** Atomic force microscopy (AFM) images of the gate Au before (up) and after (down) deposition of 1.3-nm HfO₂. Scale bar: 5 μm. **j** Height histograms of samples **(i)**.

The dielectric breakdown characteristics of as-deposited HfO₂ layers are also investigated based on the MIM structure (Fig. 2e and Supplementary Fig. 8b). Figure 2e shows leakage currents as a function of applied bias voltages for three typical dielectric thicknesses, i.e., 1.3, 2.6, and 3.9 nm. Note that both Fowler–Nordheim (FN) tunneling and direct tunneling could contribute to the leakage currents in MIM structure. The experimentally observed suppression of leakage current is consistent with the large bandgap of the dielectric with minimal oxygen vacancies and the oxygen-rich surface, which effectively reduces wavefunction overlap at the contact–dielectric interface. In contrast, similar measurements on the 1.3-nm-HfO₂ deposited via a conventional one-step-oxidation ALD process exhibit a direct tunneling behavior even under small bias voltages at ~mV scale (Supplementary Fig. 9). The hard breakdown voltage for the 1.3-nm-HfO₂ deposited via MOALD is 2.9 V, corresponding to an electrical breakdown field of ~22.3 MV/cm. Note that the leakage current of all samples consistently remains a level below the low-power threshold of 0.015 A/

cm² till breakdown. This compelling observation underscores the robust performance and reliability of our samples, highlighting their suitability for various applications particularly in the low-power scheme. In case of 1.3-nm-HfO₂ dielectrics, the leakage at 4 MV/cm gating maintains <10^{−6} A/cm² (approaching detection limit), a value significantly approaching the dynamic random-access memory (DRAM) limit^{15,25,26} (Fig. 2f). The current-time curves of the 1.3-nm-HfO₂ dielectric (Supplementary Fig. 10) also demonstrate electrical stability. These results suggest the substantial advantages of ultrathin HfO₂ dielectrics in gate-first technology for low-power applications.

Based on the above results, we benchmark the performance of our HfO₂ dielectrics in comparison with existing up-to-date EOT dielectrics such as HfO₂, SrTiO₃, CaF₂, and Bi₂SiO₅ (Fig. 2g and Supplementary Table 7)^{10,13–19,27–33}. It can be seen that MOALD ultrathin HfO₂ dielectrics exhibit one of the highest dielectric constants at the same scale of dielectric physical thickness. In Fig. 2h, It also show one of the lowest recorded EOT and electrical breakdown field strengths among all

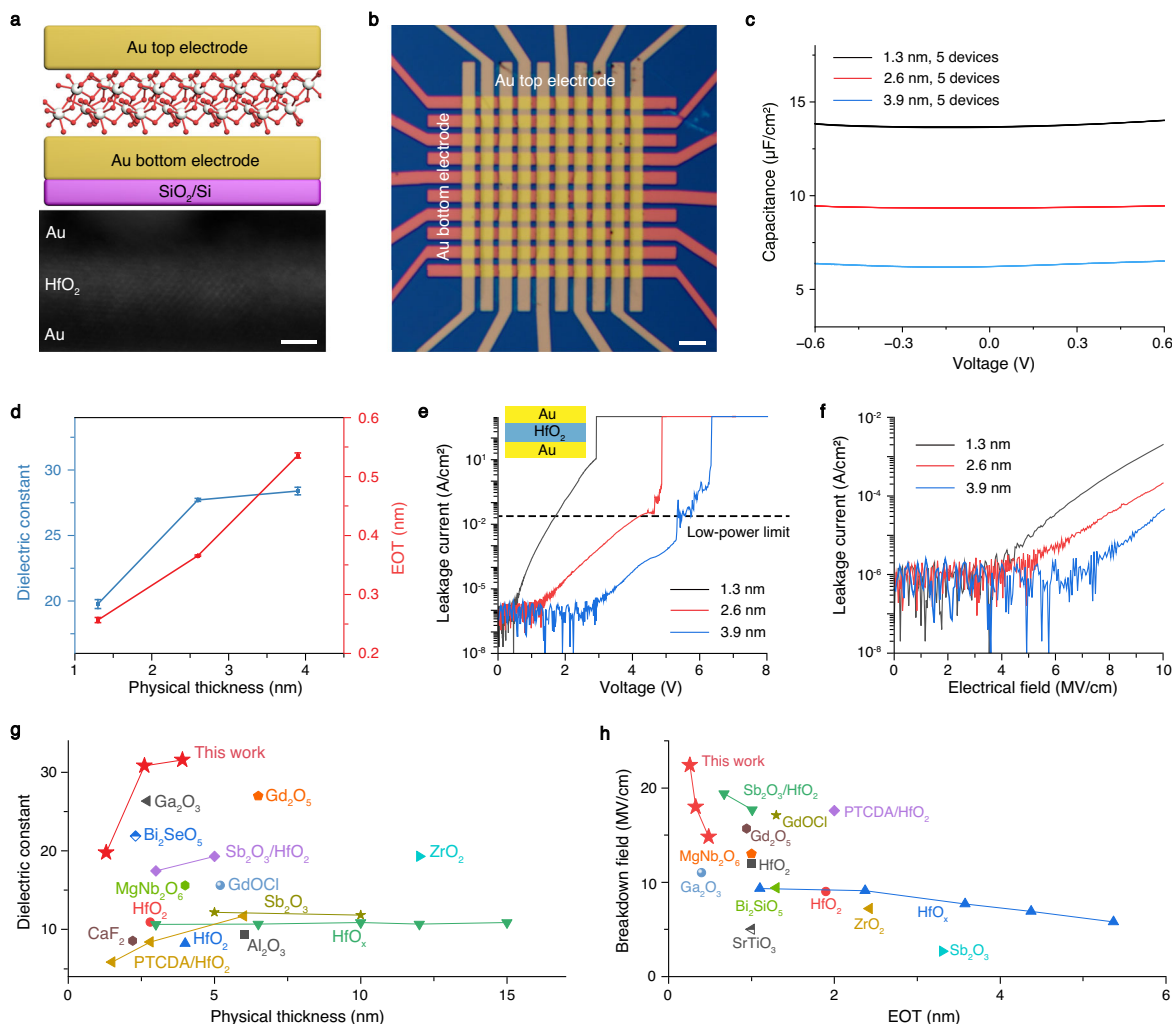


Fig. 2 | Dielectric properties of the ultrathin MOALD ultrathin HfO_2 dielectrics. **a** Schematic illustration and CS-TEM image of metal-insulator-metal (MIM) cell. Scale bar, 1 nm. **b** Optical images of ultrathin HfO_2 MIM arrays. Scale bar, 20 μm . **c** Electrical capacitance measurement of 1.3-, 2.6-, and 3.9-nm thick HfO_2 films. **d** EOT and κ of the as-deposited HfO_2 dielectrics with layer thicknesses varying from 1.3 to 3.9 nm. The error bar is calculated with standard deviation. **e** Bias-

voltage-dependent leakage current density and **f** that within 10 MV/cm in Au/ HfO_2 /Au MIM structure with HfO_2 thicknesses varying from 1.3 to 3.9 nm. **g** Comparison of the breakdown field with respect to physical thickness of different dielectric materials^{10,13–19,27–33}. **h** Benchmark for breakdown field versus EOT of different dielectric materials^{10,13–19,27–33,38}.

reported dielectric films^{1,34}. Therefore, our MOALD ultrathin HfO_2 dielectrics are promising for advanced transistors to achieve heightened carrier densities under equivalent effective gate voltages while simultaneously maintaining minimal current leakages.

2D FETs employing with 1.3-nm- HfO_2 gate dielectrics

2D FETs are considered a promising solution to future chips in the post-Moore era and usher in the ångström technology node^{4,8,35,36}. Aiming for such 2D FETs, both channels and high- κ dielectrics are required to be ultimately scaled. In this work, we further investigate such possibilities by implementing the 2.5-Å EOT high- κ dielectrics into the 2D transistor technology. Transferred monolayer MoS_2 is employed as the channel and the device fabrications are accomplished by gate-first technology in this part (Fig. 3a, Materials and methods).

Gate-first MoS_2 FETs are fabricated with varied HfO_2 thicknesses but same channel width/length (W/L) of 5/2 μm (Supplementary Fig. 11). As shown in Fig. 3b, a reduction in the HfO_2 thickness leads to decreasing gate control voltage, suggesting an enhancement in gate controllability for thinner dielectrics. Figure 3c and Supplementary Fig. 12 show the transfer curve of a typical 2.5-Å-EOT 2D FET with a 2- μm channel length, featuring with a fundamental thermionic limit of the

subthreshold slope (SS) of 70 mV/dec (Fig. 3e), a substantial on-off ratio of $\sim 10^8$ and a negligible hysteresis (~ 1 mV) at a sweep rate of 0.1 V/s, just as expected. The output characteristics are shown in Fig. 3d, and the on-state current density (I_{on}) exceeds 58 $\mu\text{A}/\mu\text{m}$ at gate voltage (V_{gs}) = 1 V and a source-drain bias of 0.5 V. The transfer curves of 105 MoS_2 FETs are shown in Fig. 3f. These devices exhibit small device-to-device variations and superior performance in terms of SS, on/off ratio, and I_{on} (Fig. 3f). The average SS is ~ 72 mV/dec , meeting the requirement of the 2037 criteria for future FETs (2022 IRDS). Moreover, these devices show a high average on/off ratio of 4×10^8 and a high I_{on} of 18 $\mu\text{A}/\mu\text{m}$ at 0.1 V source-drain bias. These excellent device performances highlight the effectiveness of the high- κ dielectrics on gate controllability.

Note that, due to the effect of vdW gap, quantum capacitance, and contact resistance, the effective capacitance in transistors is little less than the oxide capacitance. Thus, we re-estimated the transistor capacitance and CET to confirm this judgment, as shown in the Fig. 3h–k and Supplementary Figs. 13–15. From these measurements, we report an average CET of 0.34 ± 0.01 nm for 1.3-nm HfO_2 dielectrics, 0.47 ± 0.03 nm for 2.6-nm HfO_2 dielectrics, 0.66 ± 0.03 nm for 3.9-nm HfO_2 dielectrics, respectively. Short-channel transistors at

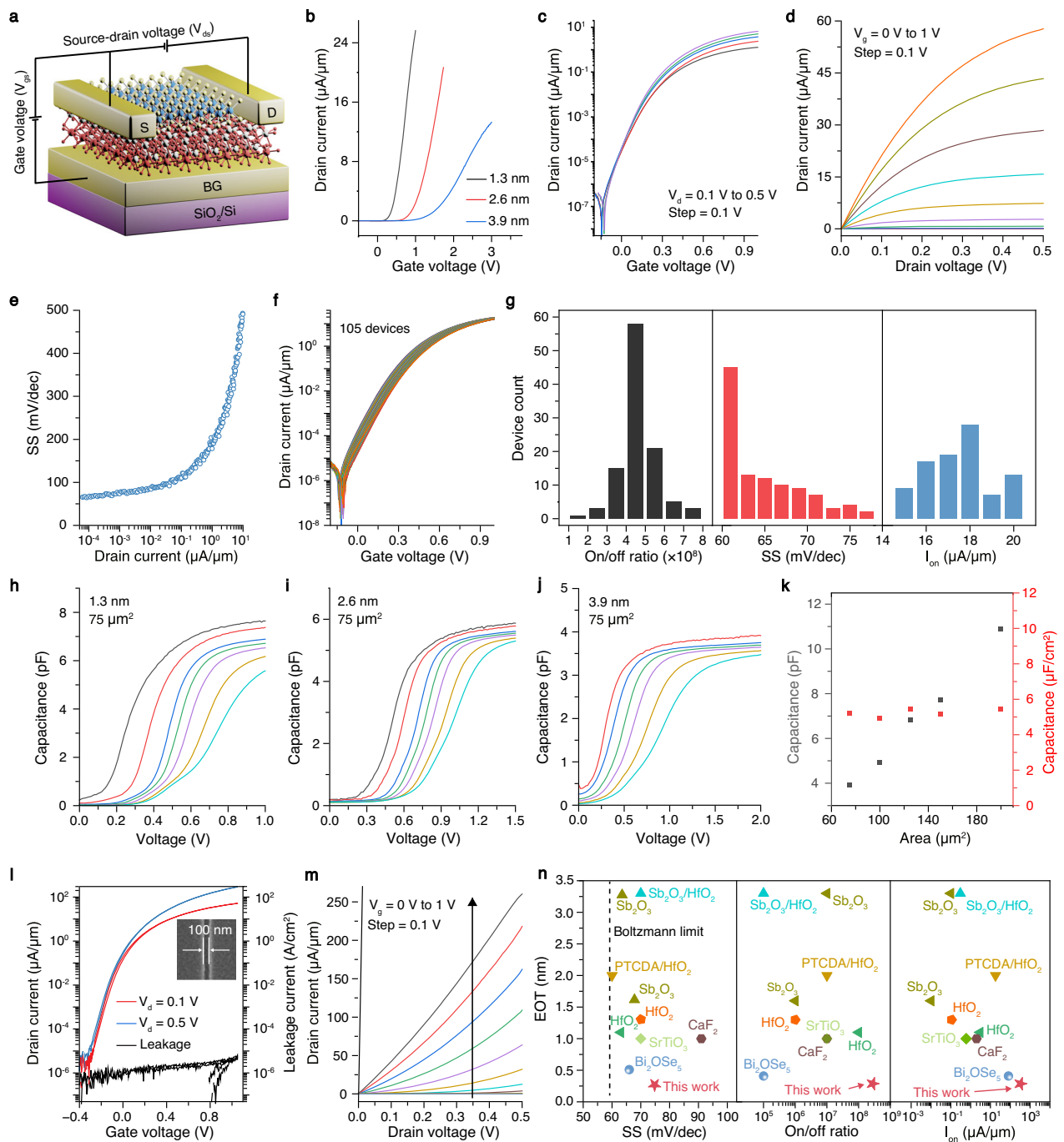


Fig. 3 | Electrical performances of MoS₂ field-effect transistors (FETs) with 2.5-Å-EOT HfO₂ dielectric layer. **a** Schematic illustration of monolayer MoS₂ gate-first field-effect transistors. V_{ds} drain-source voltage, V_{gs} gate voltage, S source, D drain. **b** Transfer curves of MoS₂ FETs with 1.3-, 2.6-, and 3.9-nm HfO₂ dielectrics in transistors with width/length (W/L) of 5/2 μm . **c** Transfer curves and **d** output curves of a MoS₂ FET with 1.3-nm HfO₂ dielectrics. **e** Corresponding subthreshold swing curve with 0.1 V source-drain bias. **f** Transfer characteristics of 105 devices with 1.3-nm HfO₂ dielectrics. **g** The corresponding statistical distribution of on/off ratio, SS,

and on-state current. C-V measurement in MoS₂ transistors with **h** 1.3 nm, **i** 2.6 nm, and **j** 3.9 nm HfO₂ dielectrics. **k** Measured capacitance and unit-area capacitance in MoS₂ transistors with 1.3 nm HfO₂ dielectrics. **l** Transfer curve and **m** output curve of shorter 100-nm-length MoS₂ transistors. The inset shows the SEM image of a typical device. **n** Benchmarks of on/off ratio, SS and on-state current against EOT achieved from our MoS₂ FETs in comparison with contemporary sub-3.5-nm-EOT MoS₂ devices.

L_{ch} = 100 nm are fabricated. Remarkably, the gate leakage at V_{gs} = 1 V remains below 10^{-5} $\mu\text{A}/\mu\text{m}$, which is comparable to those previous state-of-the-art devices^{15,17,18}. The sweeping measurements (Fig. 3l) also indicate an exceedingly small hysteresis (<10 mV) and low SS value (70 mV/dec, Supplementary Fig. 16). A high I_{on} of 260 $\mu\text{A}/\mu\text{m}$ at 0.5 V

source-drain voltage and 1 V gate voltage, which exceeds the most of MoS₂ FETs with advanced dielectrics. (Fig. 3m)

We also summarize the key performance metrics in 2D FETs incorporated with sub-3.5-nm-EOT dielectrics in Fig. 3n, with respect to the on-state current, on/off ratio, and SS which are pivotal indicators

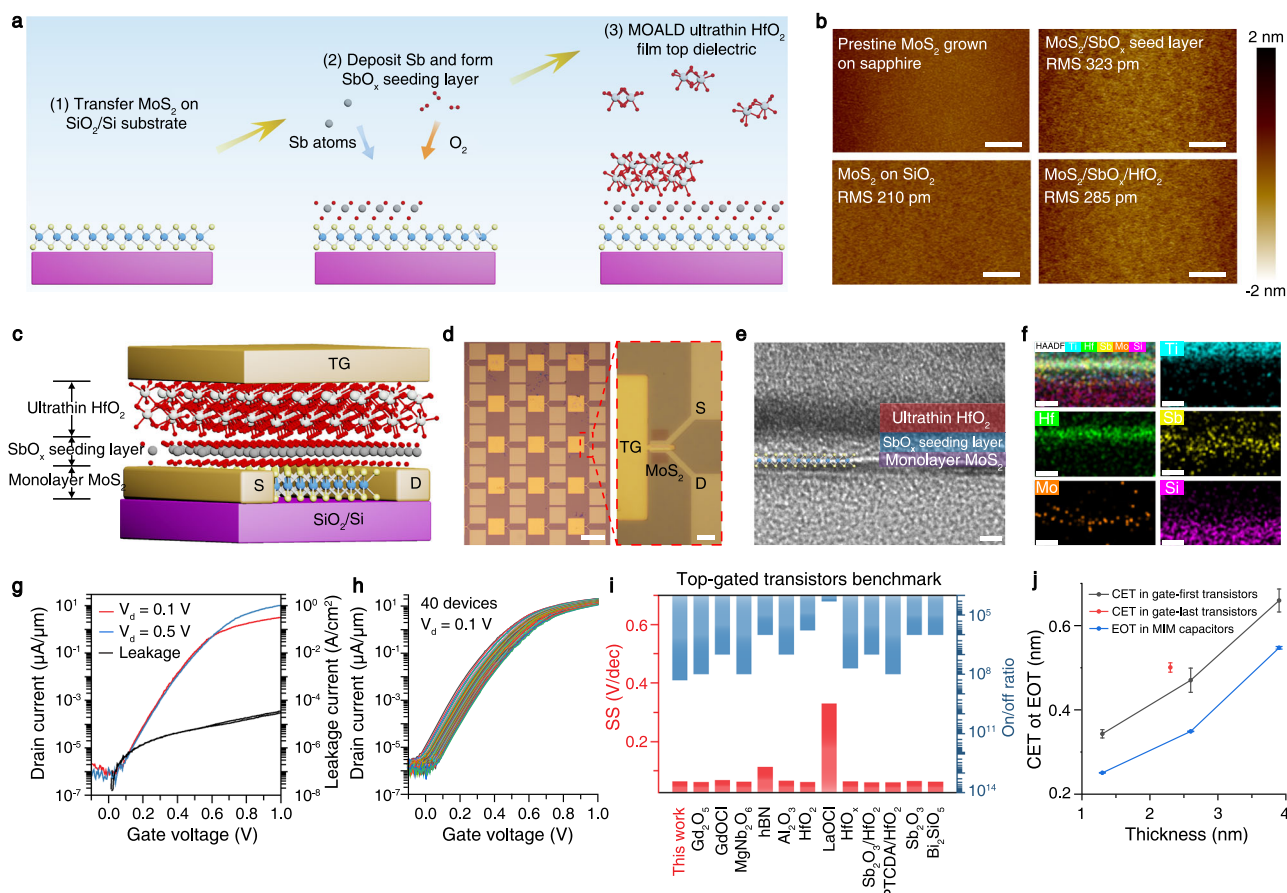


Fig. 4 | Gate-last 2D MoS₂ transistors for sub-1-V operation. **a** The fabrication process of gate-last MoS₂ FETs. After transferring MoS₂ to SiO₂ substrate, SbO_x seeding layer is necessary for preparation of depositing MOALD HfO₂. **b** AFM tomography of sample surface for every step, scale bar: 2 μ m. **c** Schematic illustration and **d** Optical images of the obtained MoS₂ gate-last transistors structure, scale bar: 100 μ m for devices array and 10 μ m for enlarged area. **e** CS-TEM and **f** High-angle annular dark-field scanning transmission electron microscopy

(HAADF-STEM) mapping of one typical MoS₂ top-gate transistors, scale bar: 1 nm. **g** Transfer curves of gate-last MoS₂ FETs with ultrathin dielectrics. **h** Transfer characteristics of 40 devices with source-drain voltage of 0.1 V. **i** Benchmarks of on/off ratio and subthreshold slope (SS) achieved from our gate-last MoS₂ FETs in comparison with other reported gate-last 2D FETs^{10,14,17,19,28,30–33,38,39}. **j** Comparison of CET and EOT value in gate-first, gate-last transistors and MIM capacitors with various thickness dielectrics. The error bar is calculated with standard deviation.

for the operational efficiency and device applicability. From this benchmark analysis, our 2D FETs exhibit obvious performance superiority in terms of the aforementioned metrics. Such performance superiority underscores the potential of 2.5-Å EOT dielectrics to not only the current silicon-based semiconductor technology but also a variety of new post-Moore technologies.

Gate-last 2D MoS₂ transistors for sub-1-V operation

For all transistors, including silicon-based CMOS and thin-film transistors (TFTs), gate-last or gate-all-around architectures are commonly adopted. Therefore, it is necessary to evaluate gate-last 2D MoS₂ transistors with ultrathin MOALD HfO₂ dielectrics^{4,5}. Due to the lack of dangling bonds in the MoS₂ surface, it is generally difficult to deposit ultrathin HfO₂ film by ALD (Supplementary Fig. 17). Here, we adopt an alternative approach. Firstly, a 1-nm-thick antimony (Sb) film is deposited on the MoS₂ surface through electron-beam evaporation as the seeding layer. Such a thin Sb film will be oxidized to SbO_x after 10 h of exposure to the air. Then we can deposit MOALD ultrathin HfO₂ film (Fig. 4a). The quality of MoS₂ after deposition is not degraded, as shown in the Raman and PL spectra (Supplementary Fig. 18). AFM is also conducted to measure the surface of every step (Fig. 4b). The schematic illustration and optical photos of the obtained gate-last MoS₂ transistors is shown in the Fig. 4c, d. The total thickness of the device is only ~3 nm, including 0.7 nm MoS₂, 1 nm SbO_x seeding layer

and 1.3 nm HfO₂, which is confirmed with CS-TEM and STEM-EDS mapping (Fig. 4e, f).

Figure 4g shows the transfer curve of an as-fabricated gate-last MoS₂ transistor, revealing a high on/off ratio of 10⁸ and average SS of 71 mV/dec. The drain current saturates at gate voltage of ~0.7 V, which is closer to the requirement of IRDS 2024. Note that the V_{th} is more positive in gate-last transistors than that in gate-first transistors. We attribute this shift primarily to the incorporation of oxygen doping within the MoS₂ channel during the fabrication of the gate-last stacks. This doping introduces n-type carriers, which effectively shifts the turn-on voltage in the positive direction⁷. The output curves of this device is shown in Supplementary Fig. 19, illustrating a high current density of 15 μ A/ μ m at 0.5 V source bias, which is slightly smaller than that in gate-first transistors. The reason may come from the small contact resistance variation³⁷ or gate controllability difference between 0.34 nm CET for gate-first devices and 0.50 nm CET for gate-last devices.

40 FETs with high uniformity are shown in the Fig. 4i. Compared with reported dielectrics for gate-last 2D transistors, our MOALD ultrathin HfO₂ shows some of the one of the best SS of 70 mV/dec and on/off ratio of 10⁸ (Fig. 4i)^{10,14,17,19,28,30–33,38,39}. The effective CET in ultrathin gate-last MoS₂ transistors with ultrathin 1 nm SbO_x and 1.3 nm HfO₂ dielectrics is also measured (Supplementary Fig. 20), demonstrating a small CET value of 0.50 nm and corresponding dielectric

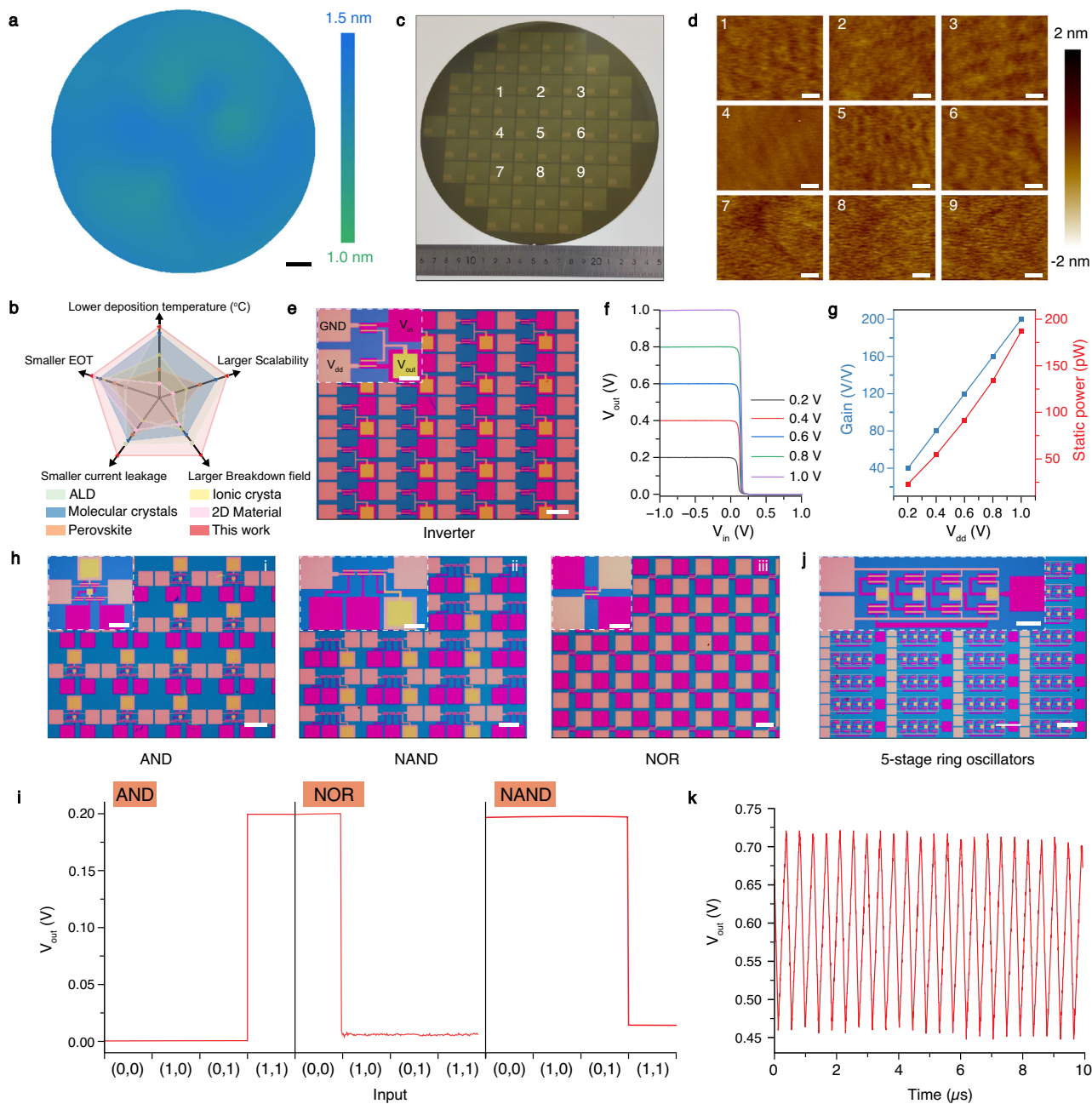


Fig. 5 | MoS₂ FETs and circuits at 2.5-Å-EOT and 8-inch wafer scale. a Thickness mapping of 8-inch 1.3-nm HfO₂ wafer, scale bar 2 cm. **b** A radar comparison between our and other dielectrics in terms of EOT, current leakage, breakdown field, scalability, and deposition temperature. **c** Photos of 8-inch MoS₂ FETs and circuits. **d** AFM tomography of HfO₂ surface in nine different positions in (c), scale bar: 2 μm. **e** Optical images of the inverter at small (scale bar: 50 μm) and large magnification (scale bar: 100 μm). **f** Output curve, **g** voltage gain and static power of

an inverter at supply voltage from 0.2 to 1.0 V. V_{dd} drain voltage, V_{in} input voltage, V_{out} output voltage. **h** Optical images at small (scale bar: 50 μm) and large magnification (scale bar: 100 μm) and **i** output characteristics of logic AND, NAND, and NOR devices at a supply voltage of 0.2 V. **j** Optical images at small (scale bar: 50 μm) and large magnification (scale bar: 200 μm) and **k** output curve of a five-stage ring oscillators (ROs) at supply voltage of 1.0 V. The output frequency is 2.35 MHz. None of the optical images in this figure uses false colors.

constant of 17.9. Comparing the EOT in the MIM structure and the CET in the MIS gate-first devices, the CET in gate-last devices is slightly larger (Fig. 4j), while still fulfilling the requirement of dielectrics in the future advanced transistors and chips.

8-inch wafer-scale 2D circuits with 1.3-nm-HfO₂

To validate the scalability of the MOALD ultrathin HfO₂ dielectrics, we deposit 1.3-nm-HfO₂ dielectric layer on the 8-inch SiO₂/Si wafer. The film thickness is accessed by ellipsometer (Fig. 5a), confirming the good uniformity with a minimal margin of error of 200 pm. Compared

with previously reported sub-1-nm-EOT dielectrics of SrTiO₃, Bi₂SeO₅, Bi₂SiO₅, Sb₂O₃, CaF₂, and HfO₂, our MOALD ultrathin HfO₂ dielectrics showcase an unparalleled characteristics and superiority (Fig. 5b)^{14–18,29}. For instance, the remarkably low process temperature, e.g., 200 °C, enables seamless integration with silicon CMOS processes, in marked contrast from both perovskite (~1000 °C) and 2D dielectric materials (~750 °C)^{15,19}. Moreover, the developed aggressively low-EOT dielectrics at 8-inch wafer scale significantly surpasses the limitations of other sub-1-nm-EOT dielectrics with a maximum wafer size of 4 inch. The 2.5-Å EOT HfO₂ dielectrics also enable remarkable gate control

capability, a high breakdown electric field, and minimal leakage current.

We further demonstrate 8-inch wafer-scale fabrication of monolayer MoS₂-based circuits with 1.3-nm-HfO₂ dielectrics (Fig. 5c). More details on the fabrication process are included in Methods. Note that the 8-inch wafer-scale monolayer MoS₂ is grown by our recently developed technique⁴⁰ (Supplementary Fig. 21). Figure 5d shows flat tomography in nine different areas of HfO₂ film. In Fig. 5e–g and Supplementary Figs. 22, 23, we present the inverter measurement in this 8-inch wafer. It shows a high gain of 40 and 205 at V_{dd} of 0.2 and 1.0 V, respectively. At the same time, the static power is only 22 pW at V_{dd} of 0.2 V. This diminutive power supply is significantly smaller than that of devices fabricated with other sub-1-nm-EOT dielectric layers, owing to the low EOT coupled with its good gate tunability. Besides, other elemental components of chips, including inverters and AND, NAND and NOR logic gates in a single pixel are fabricated, as presented in Fig. 5h, i. All AND, NAND and NOR logic gates function correctly and exhibit compatibility with the same low supply voltage of 0.2 V. Five-stage ROs could also work smoothly at V_{in} of 0.4 V with an output frequency of 2.35 MHz (Fig. 5k). These demonstrations further shed lights on applying such ultra-low-EOT high- κ dielectrics in emerging large-scale 2D electronics targeting to further scaling down and low-power consumptions⁵.

Sub-5-Å-EOT HfO₂ dielectric technology is developed at a maximum wafer size of 8-inch for both gate-first and gate-last technology. The achieved ultrathin HfO₂ film holds an ultra-low EOT of 2.5 Å, in conjunction with a low leakage current of 10^{−6} A/cm² at 1 V bias voltage, and possesses a robust breakdown electric field of ~22.3 MV/cm. We also demonstrate 8-inch wafer-scale 2D FETs and logic circuits employing 1.3-nm-HfO₂ dielectrics with remarkable electrical performances. Our results not only offer valuable insights into further scaling of EOT, approaching the physical limits of dielectric thickness, but also represent a promising step towards the development of next-generation electronic devices.

Methods

Synthesis of ultrathin HfO₂ film

First, we deposit 10 nm Au as the metal gate layer via E-beam evaporation on the cleaned 300 nm SiO₂/Si substrate. Then, 0.5-nm-Ti is deposited on metal gates as the seeding layer for ALD of HfO₂ dielectric layer. The ALD process is carried out in the Picosun R-200 Adv system. For the normal (one-step) ALD process, Tetraakis(ethylmethylamino)hafnium (TEMAH) and O₃ serve as Hf sources and oxygen sources, respectively. The pulse (purge) time are 1.6 (25) s for TEMAH and 1.6 (30) s for O₃. The carrier gas is pure N₂ with a flow rate of 150 sccm for both TEMAH and O₃. The TEMAH source is heated up to 125 °C during the deposition. For the MOALD process, the deposition condition for the Hf source remains the same, while the oxygen sources are alternated between successive O₃ and O₂ plasmas. The pulse and purge times of O₃ and O₂ plasma are 1.6 and 25 s, respectively, with the O₂ plasma operating at a power of 1500 W. Post-testing revealed film physical thickness of 1.3, 2.6, and 3.9 nm were achieved after 10, 20, and 30 cycles, respectively.

Sample characterizations

Raman and photoluminescence spectra are collected in Horiba Lab-Ram HR Evolution with a laser excitation wavelength of 532 nm and a power of 2 mW. Film thickness is measured by spectropic Mueller matrix ellipsometer ME-mapping. Scanning electron microscope images and cross-sectional STEM-HAADF images were obtained by Regulus 8100 and Thermo Fisher Spectra 300, respectively. AFM are performed by Oxford Instruments Asylum Research Cypher S. X-ray photoelectron spectroscopy measurements are conducted with the Thermo Fisher ESCALAB XI+.

Device fabrications and measurements

MoS₂ FETs and logic gates are fabricated via standard microfabrication processes. Fabrications involve UV lithography, reactive ion etching and electron-beam evaporation to define device structures, etch MoS₂ and deposit metal gate or contact, respectively. An 8-inch monolayer MoS₂ film is transferred from sapphire to high- κ /metal gate by the PMMA-assisted method. For contact metal, Au/Ti/Au (3/2/30 nm) are deposited in vacuum at a base pressure of $\sim 2 \times 10^{-7}$ torr with a rate of 0.01 Å·s^{−1}. All electrical measurements are performed in the Lake Shore probe station under 10^{−6} torr with a Keysight semiconductor analyzer B1500 equipped with high-resolution modules at room temperature.

Simulations

A “melt-quench” process and classical molecular dynamics (MD) are employed to generate initial amorphous structures⁴¹. Here, the initial HfO₂ amorphous systems are generated using this melt-quench method with MD implemented in the LAMMPS software package⁴². The interatomic potential is considered as the charge-optimized many-body potential (COMB)⁴³. The interatomic potential originally used a core-shell model with ionic charge +4 for Hf and −2 for O, respectively. The crystalline slab of HfO₂ is made up of 4 × 5 × 4 supercells of P42/nmc HfO₂ (a = b = 3.587 Å and c = 5.227 Å) with (100) plane facing upward. In this process, the crystalline systems are first heated up to 5000 °C within 10 ps, held at this temperature for 50 ps, and then quenched to 300 K within 40 ps and again held for 10 ps. After getting the amorphous structures, these systems are further optimized and calculated for the electronic properties from the First-principles calculations.

First-principles calculations are mainly performed using the ABACUS package⁴⁴, a DFT software based on numerical atomic orbitals (NAO)⁴⁵. Generalized gradient approximation with the Perdew–Burke–Ernzerhof (PBE) functional and norm-conserving Vanderbilt (ONCV) pseudopotentials from SG15 set^{46,47} is employed to describe the exchange and correlation effects of the electrons. The numerical orbital sets adopt 4s2p2d2f1g for Hf and 2s2p1d for O⁴⁸. For structure optimizations and densities of states calculation, the Brillouin zone is sampled with a Gamma-centered 2 × 3 × 1 mesh. The energy cutoff for the wave functions is set to 100 Ry.

Data availability

Relevant data supporting the key findings of this study are available within the article and the Supplementary Information file. All raw data generated during the current study are available from the corresponding authors upon request.

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Author contributions

G.Z. and N.L. co-supervised the project. S.Z. performed the HfO_2 film growth, device fabrication, characterizations, and electrical measurements. T.Z., H.G., and L.X. performed DFT calculations. X.L., T.L., and C.C. assisted with the device fabrication and electrical measurements. H.Y., S.W., L.H., and Y.Z. contributed to the 8-inch MoS_2 growth and characterizations. S.W., D.Z., L.B., S.L., C.S., W.Y., L.D., D.S., X.T., and Y.C. were involved in data analysis. S.Z., N.L., and G.Z. wrote, and all authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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